Smart Plant Mood Monitor

Arduino-Based Soil & Light Sensing with OLED Display

Software: Cirkit Designer | Platform: Arduino UNO | Language: C++ (Arduino)

1. Project Overview

The Smart Plant Mood Monitor is an Arduino-based IoT project that simulates a plant expressing its environmental needs through a personality-driven display. Two sensors continuously measure soil moisture and ambient light levels. Based on the combined sensor readings, the system selects one of four mood states and communicates it via an OLED text display — giving the plant a voice to report whether it is thriving, thirsty, cold, or sleepy.

Project Title	Smart Plant Mood Monitor
Software	Cirkit Designer
Microcontroller	Arduino UNO (ATmega328P)
Programming Language	C++ (Arduino Framework)
Sensors Used	SparkFun Soil Moisture Sensor + Photoresistor (LDR) Module
Display	OLED 128×64 I2C Monochrome (SSD1306, address 0x3C)
Libraries Required	Wire.h, Adafruit_GFX.h, Adafruit_SSD1306.h
Mood States	Cold & Thirsty / Water Me Please / Sleepy / I Feel Great
Update Interval	Every 2 seconds (2000 ms delay)

2. Components Required

SL.NO	Component	Specification / Role	Pin / Address	Qty
1	Arduino UNO	ATmega328P, 5V, 14 digital + 6 analog I/O	Central MCU	1
2	SparkFun Soil Moisture Sensor	Analog output, measures soil conductivity	A0	1
3	Photoresistor (LDR) Module	Analog output, measures ambient light	A1	1
4	OLED 128×64 I2C Display	SSD1306 controller, monochrome, 3.3–5V	SDA/SCL (I2C)	1
5	Resistors (200 Ω)	Current limiting for sensor signal lines	—	2

6	Breadboard	Full-size, for component interconnection	—	1
7	Jumper Wires	Male-to-Male for all connections	—	Many
8	USB Cable	Type-A to Type-B, powers Arduino via PC	—	1

3. Circuit Connections

Soil Sensor – VCC	→ 5V rail (breadboard)
Soil Sensor – GND	→ GND rail (breadboard)
Soil Sensor – SIG	→ Arduino A0 (via 200Ω resistor)
LDR Module – VCC	→ 5V rail
LDR Module – GND	→ GND rail
LDR Module – AO	→ Arduino A1 (via 200Ω resistor)
OLED – VDD	→ 3.3V or 5V rail
OLED – GND	→ GND rail
OLED – SDA	→ Arduino A4 (Hardware I2C SDA)
OLED – SCL	→ Arduino A5 (Hardware I2C SCL)

4. Mood Decision Logic

The Arduino evaluates both sensor readings in real time and maps them to one of four moods, each communicated through a distinct message on the OLED display:

Soil Moisture	Light Level	OLED Message
< 300 (Dry)	< 300 (Dark)	'Cold & Thirsty!'
< 300 (Dry)	≥ 300 (Bright)	'Water Me Please!'
≥ 300 (Moist)	< 300 (Dark)	'Sleepy zzz...'
≥ 300 (Moist)	≥ 300 (Bright)	'I Feel Great!'

5. Simulation Screenshots – Mood States

The following four screenshots, captured from Cirkit Designer, show the circuit for each plant mood state based on the sensor input combination. The OLED display shows the corresponding mood message.

Case 1 – Cold & Thirsty (Soil < 300, Light < 300)

Both soil moisture and light levels are low. The plant is cold and thirsty. The OLED displays 'Cold & Thirsty!'

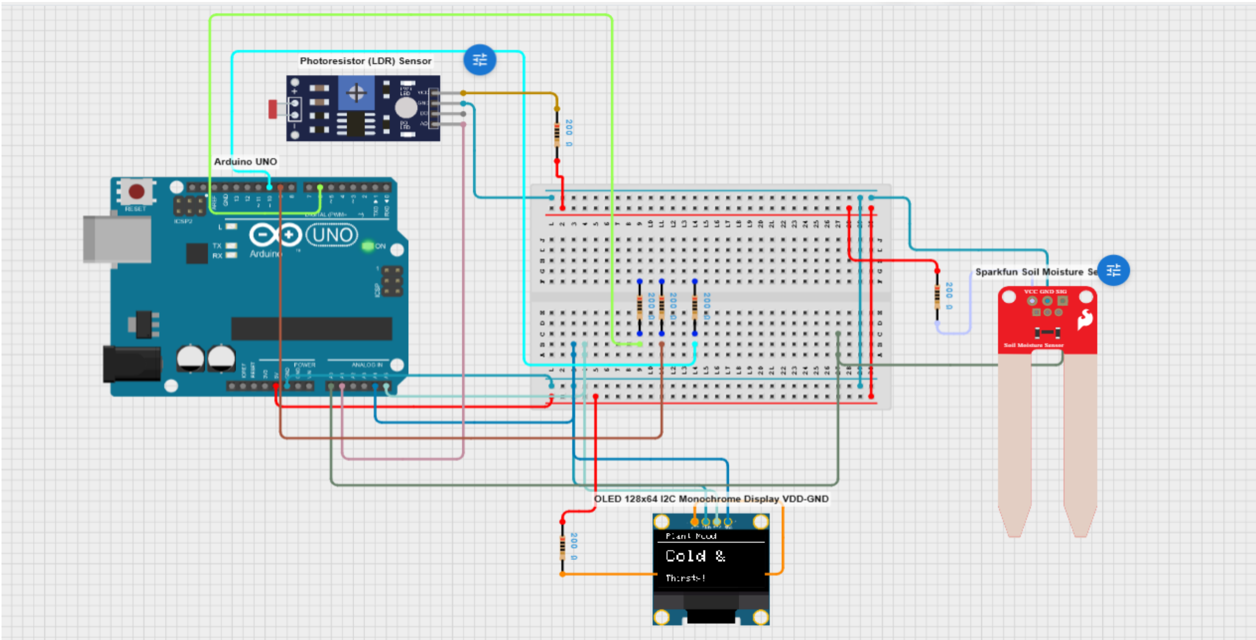

Figure 1: Case 1 — Soil dry + Dark environment → 'Cold & Thirsty!'

Case 2 – Water Me Please (Soil < 300, Light ≥ 300)

Soil moisture is low but light is adequate. The plant needs water. The OLED displays 'Water Me Please!'

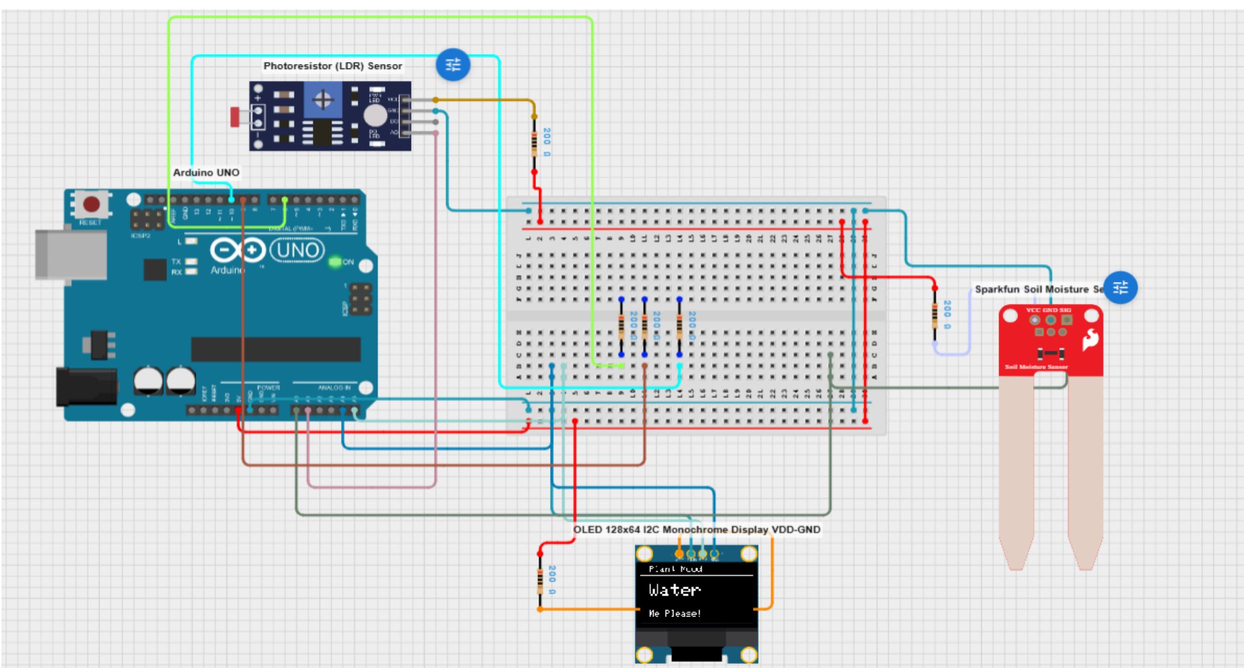

Figure 2: Case 2 — Soil dry + Bright light → 'Water Me Please!'

Case 3 – Sleepy (Soil ≥ 300, Light < 300)

Soil moisture is sufficient but light is dim — it's dark. The plant is resting. The OLED displays 'Sleepy zzz...'

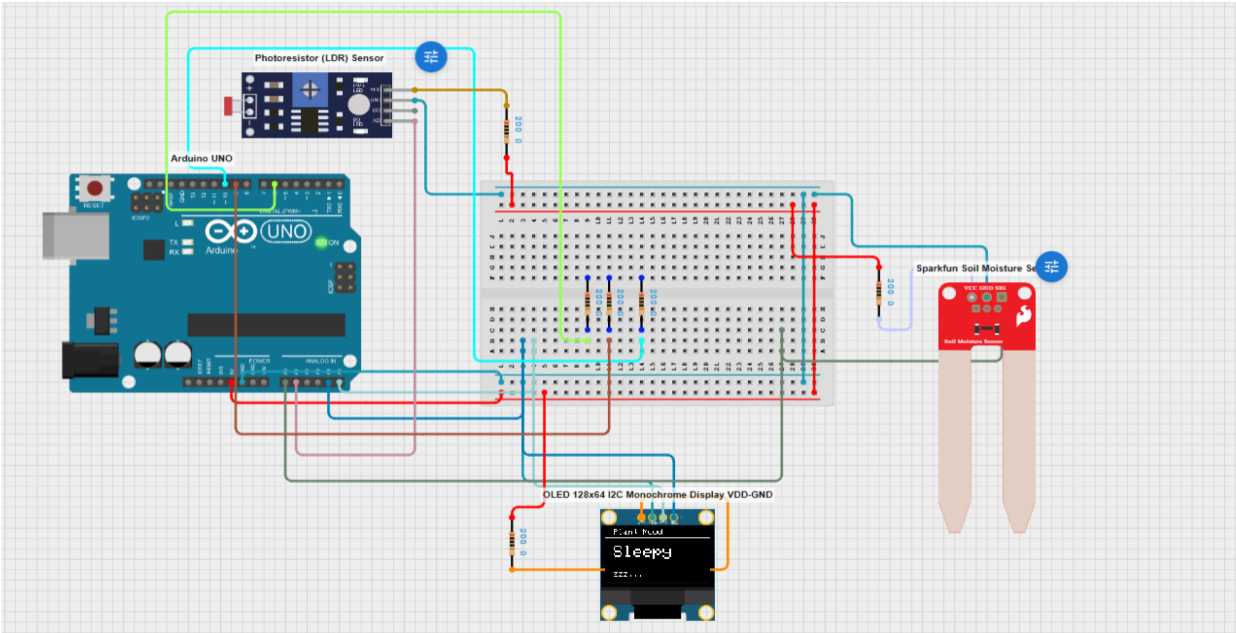

Figure 3: Case 3 — Soil moist + Dark environment → 'Sleepy zzz...'

Case 4 – I Feel Great (Soil ≥ 300, Light ≥ 300)

Both sensors report good conditions. The plant is happy and healthy. The OLED displays 'I Feel Great! JAY'

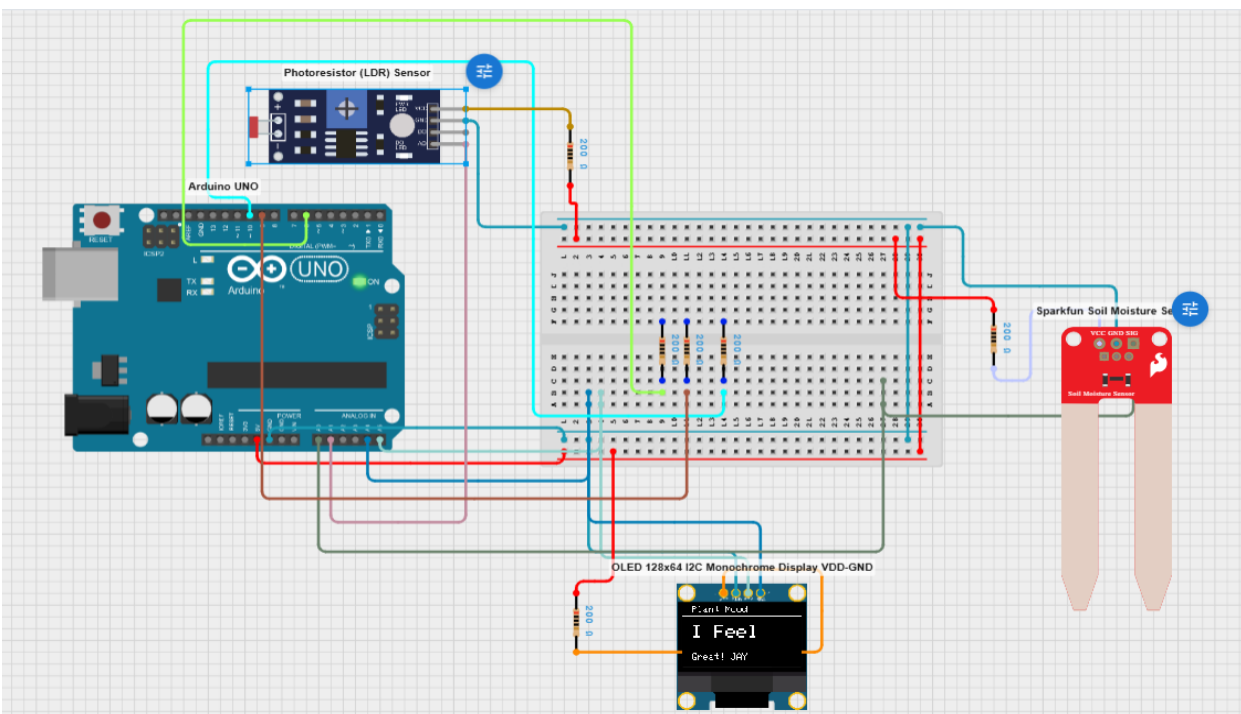

Figure 4: Case 4 — Soil moist + Bright light → 'I Feel Great!'

6. Arduino Source Code

The complete sketch is listed below. It reads both analog sensors every 2 seconds, evaluates the mood condition, and calls setMood() to update the OLED display.

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

int soilPin = A0;
int lightPin = A1;

void setup() {
  Serial.begin(9600);
  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);
  display.display();
}

void setMood(String line1, String line2) {
  display.clearDisplay();
  display.setTextSize(1);
  display.setCursor(10, 0);
  display.println("Plant Mood");
  display.drawLine(0, 10, 127, 10, SSD1306_WHITE);
  display.setTextSize(2);
  display.setCursor(10, 20);
  display.println(line1);
  display.setTextSize(1);
  display.setCursor(10, 50);
  display.println(line2);
  display.display();
}

void loop() {
  int soil = analogRead(soilPin);
  int light = analogRead(lightPin);

  if (soil < 300 && light < 300) {
    setMood("Cold &", "Thirsty!");
  } else if (soil < 300 && light >= 300) {
    setMood("Water", "Me Please!");
  } else if (soil >= 300 && light < 300) {
    setMood("Sleepy", "zzz...");
  } else {
    setMood("I Feel", "Great! JAY");
  }

  delay(2000);
}
```

7. Code Explanation

6.1 Library Includes & Display Object

- Wire.h enables I2C communication between Arduino and the OLED display.
- Adafruit_GFX.h provides the graphics primitives (text, lines, shapes).
- Adafruit_SSD1306.h is the driver library for the SSD1306 OLED controller.
- A display object is created with 128×64 resolution and I2C address 0x3C.

6.2 setup()

- Serial.begin(9600) opens the serial port for optional debug monitoring.
- display.begin() initialises the OLED and clears it before the main loop starts.

6.3 setMood() Helper Function

- Accepts two message strings as parameters.
- Clears the OLED, draws a title ('Plant Mood') and a horizontal separator line.
- Renders line1 at text size 2 (large) and line2 at text size 1 (small), then calls display.display() to push the buffer to the screen.

6.4 loop()

- analogRead(A0) reads raw soil moisture (0–1023); values below 300 indicate dry soil.
- analogRead(A1) reads ambient light via the LDR module; values below 300 indicate darkness.
- A 2-level if/else chain evaluates both readings and calls setMood() with the appropriate message.
- delay(2000) waits 2 seconds before the next reading cycle.

8. Cirkit Designer Simulation Steps

- Open Cirkit Designer (www.cirkitdesigner.com) and create a new project.
- Add an Arduino UNO from the component library to the canvas.
- Add a SparkFun Soil Moisture Sensor and connect VCC→5V, GND→GND, SIG→A0 via a 200Ω resistor.
- Add a Photoresistor (LDR) Module and connect VCC→5V, GND→GND, AO→A1 via a 200Ω resistor.
- Add an OLED 128×64 I2C display and connect VDD→5V, GND→GND, SDA→A4, SCL→A5.
- Open the Code Editor, select Arduino, and paste the sketch from Section 6.
- Add the Adafruit_GFX and Adafruit_SSD1306 libraries in the Libraries panel.
- Click Simulate — adjust the soil and light sensor sliders to cycle through all four mood cases.
- Observe the OLED message change with each 2-second update.

9. Learning Outcomes

- Reading multiple analog sensors (soil moisture + LDR) on separate Arduino analog pins.
- I2C communication protocol for interfacing the OLED SSD1306 display.
- Conditional logic (multi-variable if/else) for sensor-driven decision making.
- Modular function design (setMood) to keep loop() clean and readable.
- Hands-on simulation experience with Cirkit Designer including library management.

10. Conclusion

The Smart Plant Mood Monitor successfully demonstrates how multiple environmental sensors can be fused to drive an expressive, human-readable output. By combining soil moisture and light readings with an OLED display, the project creates an engaging, personality-driven interface that communicates plant status at a glance. The Cirkit Designer simulation validates all four operational scenarios before any physical hardware is required, making it an ideal rapid-prototyping exercise for embedded IoT development.